Chapter 12 — Nitrogen and Sulfur

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Nitrogen is inert due to:

(a) Weak bond **(b)** Strong N≡N bond **(c)** Polarity **(d)** Low mass

2. N_2 bond enthalpy about:

(a) +944 kJ mol^{-1} **(b)** +100 kJ mol^{-1} **(c)** +436 kJ mol^{-1} **(d)** +242 kJ mol^{-1}

3. Ammonia acts as base by:

(a) Accepting e^- **(b)** Donating O_2- **(c)** Donating H^+ **(d)** Accepting H^+

4. Ammonium ion shape:

(a) Pyramidal **(b)** Bent **(c)** Tetrahedral **(d)** Linear

5. Lab NH_3 from ammonium salt uses:

(a) NaCl **(b)** Strong acid **(c)** Strong base **(d)** Water only

6. $NH_4Cl + Ca(OH)_2$ gives:

(a) NO_2 **(b)** N_2 **(c)** NH_3 **(d)** HNO_3

7. NO is:

(a) Colourless **(b)** Yellow **(c)** Brown **(d)** Green

8. NO_2 is:

(a) White **(b)** Brown **(c)** Colourless **(d)** Blue

9. NOx forms in engines due to:

(a) No oxygen **(b)** Low temperature **(c)** High temperature **(d)** No nitrogen

10. Photochemical smog needs:

(a) Darkness **(b)** Only rain **(c)** Sunlight **(d)** Only snow

11. PAN stands for:

(a) Polyatomic nitrate **(b)** Peroxyacyl nitrate **(c)** Primary acid nitrate **(d)** Pure ammonia nitrate

12. Catalytic converter converts NOx to:

(a) HNO_3 **(b)** NO_2 **(c)** N_2 **(d)** NH_3

13. Catalytic converter converts CO to:

(a) C **(b)** CO_2 **(c)** O_2 **(d)** CH_4

14. Nitrification converts ammonia to:

(a) N_2 **(b)** NO_2 only **(c)** Nitrates **(d)** Proteins

studysapiens.com

15. Denitrification converts nitrates to:

(a) N_2 **(b)** Proteins **(c)** NO **(d)** NH_3

16. Sulfur oxidation state in H_2S:

(a) +4 **(b)** 0 **(c)** -2 **(d)** +6

17. Sulfur oxidation state in SO_2:

(a) 0 **(b)** -2 **(c)** +4 **(d)** +6

18. Sulfur oxidation state in SO_3:

(a) 0 **(b)** +4 **(c)** -2 **(d)** +6

19. Sulfur burns in oxygen to form:

(a) SO_2 **(b)** H_2SO_4 **(c)** H_2S **(d)** SO_3 directly

20. Sulfur with iron forms:

(a) FeS **(b)** $FeCl_2$ **(c)** $FeSO_4$ **(d)** Fe_2O_3

21. Vulcanization improves:

(a) Wood **(b)** Rubber **(c)** Paper **(d)** Glass

22. Sulfur is used in:

(a) Only water **(b)** Only glass **(c)** Fertilizers **(d)** Only fuels

23. Contact process catalyst:

(a) Fe **(b)** Cu **(c)** V_2O_5 **(d)** Ni

24. SO_3 is absorbed in:

(a) Water directly **(b)** Air **(c)** Concentrated H_2SO_4 **(d)** NaCl solution

25. Oleum formula:

(a) H_2SO_4 **(b)** H_2SO_3 **(c)** H_2S_2O7 **(d)** SO_2

26. H_2SO_4 is a:

(a) Reducing agent only **(b)** Dehydrating agent **(c)** Weak base **(d)** Salt

27. H_2SO_4 is used in:

(a) Only soap **(b)** Only food **(c)** Only glass **(d)** Batteries

28. Natural NOx source:

(a) Only plastics **(b)** Lightning **(c)** Only factories **(d)** Only cars

29. Nitrogen protective blanket prevents:

(a) Oxidation **(b)** Freezing **(c)** Reduction only **(d)** Melting

30. Ammonia with HCl gives:

(a) HNO_3 **(b)** N_2 **(c)** NO **(d)** NH_4Cl

31. Nitrogen molecule has:

(a) Ionic bond **(b)** Double bond **(c)** Triple bond **(d)** Single bond

studysapiens.com

32. N_2 bond enthalpy is about:

(a) +100 kJ mol^{-1} **(b)** +944 kJ mol^{-1} **(c)** +242 kJ mol^{-1} **(d)** +436 kJ mol^{-1}

33. Ammonia acts as base by:

(a) Accepting H^+ **(b)** Donating H^+ **(c)** Accepting e^- **(d)** Donating O_2-

34. Ammonium ion shape is:

(a) Tetrahedral **(b)** Bent **(c)** Pyramidal **(d)** Linear

35. Lab preparation of NH_3 uses ammonium salt and:

(a) Water only **(b)** NaCl **(c)** Strong acid **(d)** Strong base

36. NO is:

(a) Green **(b)** Brown **(c)** Colourless **(d)** Yellow

37. NO_2 is:

(a) Colourless **(b)** Blue **(c)** White **(d)** Brown

38. NOx forms in engines due to:

(a) High temperature **(b)** No oxygen **(c)** Low temperature **(d)** No nitrogen

39. Photochemical smog needs:

(a) Darkness **(b)** Only snow **(c)** Sunlight **(d)** Only rain

40. PAN stands for:

(a) Polyatomic nitrate **(b)** Primary acid nitrate **(c)** Pure ammonia nitrate **(d)** Peroxyacyl nitrate

41. Catalytic converter converts NOx to:

(a) HNO_3 **(b)** N_2 **(c)** NO_2 **(d)** NH_3

42. Nitrification converts ammonia to:

(a) N_2 **(b)** Nitrates **(c)** Proteins **(d)** NO_2 only

43. Denitrification converts nitrates to:

(a) NH_3 **(b)** NO **(c)** N_2 **(d)** Proteins

44. Sulfur occurs as:

(a) S atoms only **(b)** S_2 chains **(c)** S_4 cubes **(d)** S_8 rings

45. Sulfur oxidation state in H_2S is:

(a) -2 **(b)** +4 **(c)** 0 **(d)** +6

46. Sulfur oxidation state in SO_2 is:

(a) 0 **(b)** +6 **(c)** +4 **(d)** -2

47. Sulfur oxidation state in SO_3 is:

(a) +4 **(b)** -2 **(c)** 0 **(d)** +6

studysapiens.com

48. Vulcanization improves:

(a) Rubber **(b)** Glass **(c)** Paper **(d)** Wood

49. Contact process catalyst is:

(a) V_2O_5 **(b)** Fe **(c)** Cu **(d)** Ni

50. Oleum formula is:

(a) H_2SO_3 **(b)** SO_2 **(c)** H_2S_2O7 **(d)** H_2SO_4

studysapiens.com

Answer Key with Brief Reasons

1. Answer: (b) Strong N≡N bond — Triple bond strength.

2. Answer: (a) +944 kJ mol⁻¹ — Book value.

3. Answer: (d) Accepting H^+ — Lone pair on N.

4. Answer: (c) Tetrahedral — NH_4^+ has four bond pairs.

5. Answer: (c) Strong base — Displaces NH_3.

6. Answer: (c) NH_3 — Lab preparation.

7. Answer: (a) Colourless — Oxide of nitrogen.

8. Answer: (b) Brown — Brown gas.

9. Answer: (c) High temperature — $N_2 + O_2$ at high T.

10. Answer: (c) Sunlight — NOx + hydrocarbons.

11. Answer: (b) Peroxyacyl nitrate — Smog component.

12. Answer: (c) N_2 — Reduction.

13. Answer: (b) CO_2 — Oxidation.

14. Answer: (c) Nitrates — Bacterial oxidation.

15. Answer: (a) N_2 — Returns N_2 to air.

16. Answer: (c) -2 — H is +1.

17. Answer: (c) +4 — O is -2.

18. Answer: (d) +6 — O is -2.

19. Answer: (a) SO_2 — Combustion.

20. Answer: (a) FeS — Sulfide formation.

21. Answer: (b) Rubber — Cross-links.

22. Answer: (c) Fertilizers — Industrial use.

23. Answer: (c) V_2O_5 — SO_2 to SO_3.

24. Answer: (c) Concentrated H_2SO_4 — Forms oleum.

25. Answer: (c) H_2S_2O7 — Disulfuric acid.

26. Answer: (b) Dehydrating agent — Removes water.

27. Answer: (d) Batteries — Lead-acid batteries.

28. Answer: (b) Lightning — High-energy fixation.

29. Answer: (a) Oxidation — Inert atmosphere.

30. Answer: (d) NH_4Cl — White fumes.

31. Answer: (c) Triple bond — N≡N.

studysapiens.com

32. Answer: (b) +944 kJ mol^{-1} — Book value.

33. Answer: (a) Accepting H^+ — Lone pair on N.

34. Answer: (a) Tetrahedral — Four bond pairs.

35. Answer: (d) Strong base — Displaces NH_3.

36. Answer: (c) Colourless — Oxide of nitrogen.

37. Answer: (d) Brown — Brown gas.

38. Answer: (a) High temperature — $N_2 + O_2$ at high T.

39. Answer: (c) Sunlight — NOx + hydrocarbons.

40. Answer: (d) Peroxyacyl nitrate — Smog component.

41. Answer: (b) N_2 — Reduction.

42. Answer: (b) Nitrates — Bacterial oxidation.

43. Answer: (c) N_2 — Returns N_2 to air.

44. Answer: (d) S_8 rings — Molecular sulfur.

45. Answer: (a) -2 — H is +1.

46. Answer: (c) +4 — O is -2.

47. Answer: (d) +6 — O is -2.

48. Answer: (a) Rubber — Sulfur cross-links.

49. Answer: (a) V_2O_5 — SO_2 to SO_3.

50. Answer: (c) H_2S_2O7 — Disulfuric acid.

study-sapiens.com